TH450A/TH550A
TH650A
TH850A
TH1050A/TS2000/TS2100

Industrial Robot

INSTRUCTION MANUAL

SAFETY MANUAL

Notice

1. Make sure that this instruction manual is delivered to the final user of SHIBAURA MACHINE's industrial robot.
2. Before operating the industrial robot, read through and completely understand this manual.
3. After reading through this manual, keep it nearby for future reference.

Rev.3
April 1, 2020

Shibaura Machine

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Preface

This manual describes the safety measures for the system robots (TH450A,TH550A,TH650A, TH850A and TH1050A) built by SHIBAURA MACHINE. You are requested to read through this manual and handle the robot, strictly observing the instructions given throughout the manual, so that you can completely understand the performance of the robot and use its functions safely over the long years to come.

This manual consists of the following sections.

Section 1 Cautions on Safety
This section deals with the important information on using the robot safely and properly.

Section 2 Locations of Warning Labels
This section describes the locations of warning labels affixed to the robot and controller.

Section 3 Safety Measures
This section describes the safety functions of the robot and controller, and safety cautions on installing and operating the robot.

Cautions on Safety

This manual contains the important information on the robot and controller to prevent injury to the operators and persons nearby, to prevent damage to assets and to ensure correct use.
Make sure that the following details (indications and symbols) are well understood before reading this manual. Always observe the information that is noted.

[Explanation of indications]

Indication	Meaning of indication
 **DANGER**	This means that "incorrect handling will lead to fatalities or serious injuries."
 **WARNING**	This means that "incorrect handling will lead to fatalities or serious injuries."
 **CAUTION**	This means that "incorrect handling may lead to personal injuries *1) or physical damage *2)."

*1) Injuries refer to injuries, burns and electric shocks, etc., which do not require hospitalization or long-term medical treatment.
*2) Physical damage refers to damages due to destruction of assets or resources.

[Explanation of symbols]

Symbol	Meaning of symbol
	This means that the action is prohibited (must not be done). The details of the actions actually prohibited are indicated with pictures or words in or near the symbol.
	This means that the action is mandatory (must be done). The details of the actions that must be done are indicated with pictures or words in or near the symbol.
	This means danger, warning or caution. The details of the actual danger, warning or caution are indicated with pictures or words in or near the symbol.

[Operation]

 **DANGER**	
 Prohibited	• During operation, NEVER enter the dangerous area of the robot. Otherwise, you will be injured seriously. • DO NOT leave in the working range any machinery or materials which will hinder the operation. If the equipment went wrong, a person nearby will be injured or involved in an accident. • Anyone other than the operator MUST NOT approach the equipment. Should he negligently touch a dangerous part of the equipment, he will get injured or involved in a serious accident. • NEVER perform an inappropriate operation which is not described in the instruction manual. Otherwise, the equipment will start by mistake, resulting in personal injury or serious accident.
 Mandatory	• If you feel even a little that you are exposed to danger or the equipment works abnormally, press the EMERGENCY stop pushbutton switch to stop the equipment. If the equipment is used as it is, you will be injured or involved in a serious accident. When this happens, ask our after-sale service agent for repair. • During operation, be sure to close the equipment cover. Should the cover be opened during operation, you will be struck by an electric shock or get injured. • Only a well-trained and qualified person is allowed to perform the operation. Should the equipment be operated improperly, it will start by mistake, causing a personal injury or serious accident. • If the equipment has malfunctioned, turn the power off, identify and remove the cause of the abnormality, maintain the peripheral equipment and completely restore the malfunctioned equipment. Then start the equipment at a low speed. If the equipment starts, leaving the abnormality, you will be involved in a serious accident.

! CAUTION	
 Prohibited	• DO NOT change the data of the system parameter file (SYSTEM.PAR). Otherwise, the robot will operate abnormally, resulting in damage or an accident.
 Mandatory	• In principle, teaching operation should be performed outside the dangerous area of the robot. If it should be performed inevitably within the dangerous area, strictly observe the following matters. (1) The teaching operation should be always performed by two (2) persons. One person performs the job and the other person watches outside the dangerous area. Also, both persons should try to prevent mis-operation with each other. (2) The operator should do the job in an attitude ready to press the EMERGENCY stop pushbutton switch at any time. Also, he should perform the job at a position from which he can evacuate immediately at the time of an emergency after confirming the robot working range and shields nearby. (3) The supervisor should keep watch on the job at a position where he can see the entire robot system and operate the EMERGENCY stop pushbutton switch at the time of an emergency. Also, he should keep anyone from entering the dangerous area. • If an abnormality has generated or the POWER LED lamp on the control panel remains off after the main power switch of the equipment was turned on, turn off the main power immediately and confirm the wiring. Otherwise, you will be struck by an electric shock or a fire will break out. • Unless the robot operates toward a designated direction at manual guide, turn off the servo power. Otherwise, the robot will be damaged or you will be involved in an accident. When this happens, call us at the after-sale service agent. • Pushbutton operations of the control panel and teach pendant should be confirmed visually. Otherwise, you will be involved in an accident due to mis-operation.

CAUTION	
 Mandatory	• Before operating the equipment, perform the following inspection. (1) Make sure that visual appearance of the robot, controller, peripheral equipment and cables is in good condition. (2) Make sure that no obstacle stands in or near the operating range of the robot and peripheral equipment. (3) Make sure that the emergency stop and other safety devices operate properly. (4) Make sure that no abnormal noise or vibration is involved in the robot operation. If the above prior inspection is skipped, the equipment will be damaged or you will be involved in an accident.
 Caution	• The speed of test operation is initially set at 20 % of the maximum robot speed. • The speed of automatic operation is initially set at 100 % of the maximum robot speed. • Once a cold start has been executed, all data saved in the controller memory, such as user program, position data, user parameter and system parameter files are initialized to the factory-set values.

[Installation and transportation]

Strictly observe the following items to use the robot safely.

 **DANGER**	
 Prohibited	• DO NOT install or operate the robot if any part is damaged or missing. Otherwise, electric shocks, fires or faults will be caused. • DO NOT install the robot at a place where it is exposed to splash of water or other fluids. Otherwise, electric shocks, fires or faults will be caused. • DO NOT place the robot near a combustible material. If it ignites due to a fault, etc., a fire will break out.
 Mandatory	• Always secure the robot with attached clamps before carrying it. Otherwise, you will be injured if the arm moves when the robot is lifted. • Wire the robot after installation. Otherwise, electric shocks or injuries will be caused. • Always use the line voltage and power capacity designated by SHIBAURA MACHINE. Otherwise, the equipment will be damaged or a fire will break out. • Always use the designated power cables. (For details, see the "Transportation and Installation Manual.") If a cable other than the designated is used, fires or faults will be caused. • Install the controller outside the dangerous area where the operator can always watch the robot movements. Otherwise, it is very dangerous should the robot start during the controller operation.
 Always ground	• Completely connect the grounding cable. Otherwise, electric shocks or fires will be caused if a fault or fault current occurs. Also, it could cause mis-operation by noise.

! CAUTION	
 Prohibited	• NEVER lift the robot by the arm 2 cover or arm 2. Otherwise, an excessive force will be exerted on the robot mechanism, resulting in damage of the robot. • For the controller, secure an ample space for ventilation. (For details, see the "Transportation and Installation Manual.") Otherwise, the controller will heat and go wrong.
 Mandatory	• When lifting the robot, lift it up slowly as the robot will tilt slightly. If it is lifted up suddenly, it will cause a very hazardous situation. • When storing the robot, secure it to the base completely. If the robot is just placed on the floor, it becomes unstable and will fall down.
 Caution	• When operating the robot after long-hour stop at a low temperature (10°C or less), be sure to perform a continuous operation at a low speed (approximately 20 % of the maximum speed) for a few minutes. Otherwise, a motor overload error may occur due to solidified grease.

[Maintenance and inspection]

Strictly observe the following items to use the robot safely.

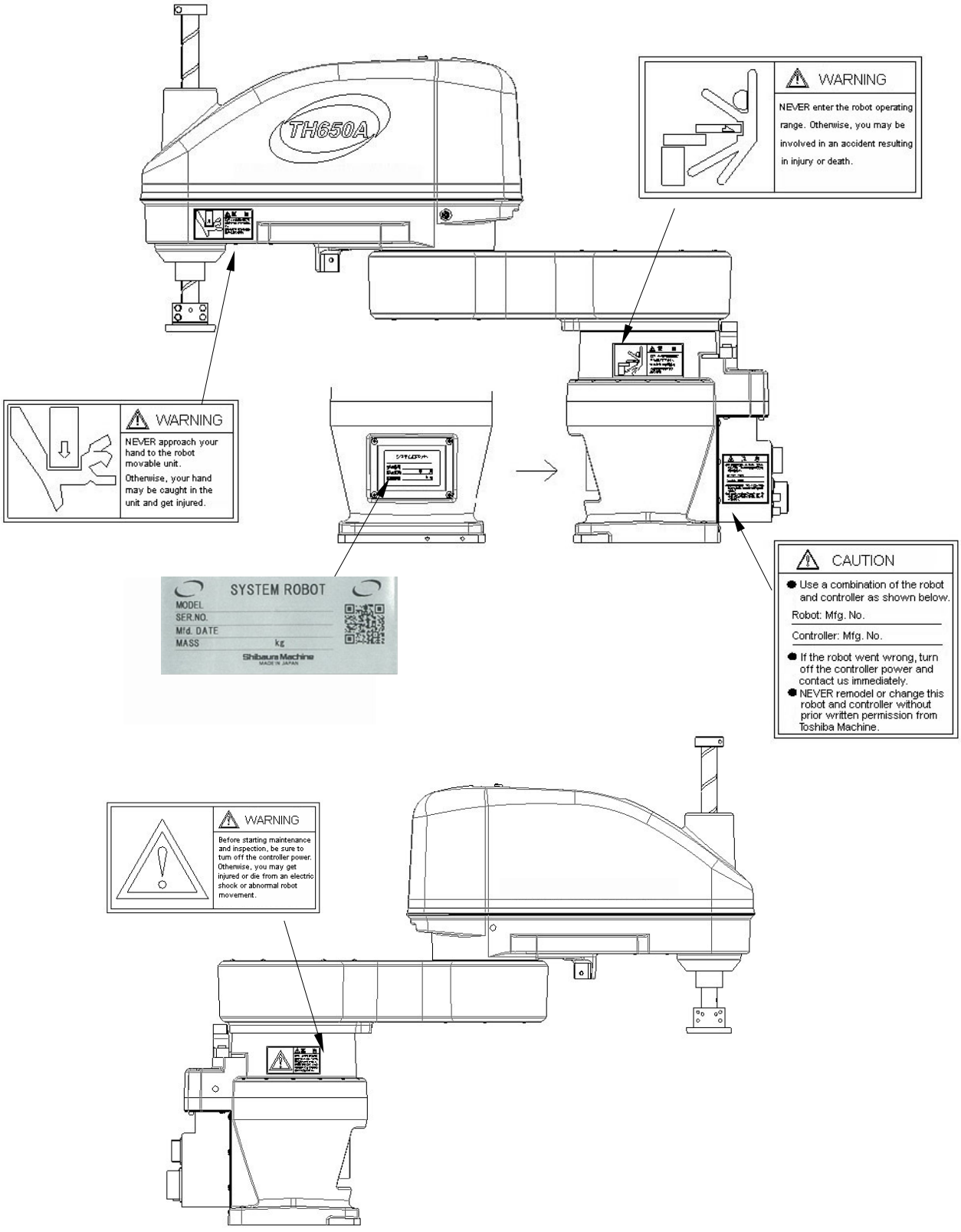
DANGER	
 Prohibited	• DO NOT incinerate, disassemble or charge the battery. Otherwise, it will rupture.
 Mandatory	• Be sure to turn off the main power of the controller before starting inspection or maintenance. • When disposing of the battery, follow the user's provided regulations.

CAUTION	
 Prohibited	• The user NEVER replace or modify parts other than those described in the instruction manual. Otherwise, the robot performance will deteriorate, or faults or accidents will be caused.
 Mandatory	• Always use the SHIBAURA MACHINE designated spare parts when replacing the parts. • Perform maintenance and inspection regularly. Otherwise, the equipment will go wrong or you will be involved in an accident.
 Caution	• The axis 4 motor of the TH450A and TH550A and TH650A robots is not provided with a brake. At servo OFF, therefore, the axis 4 may rotate due to the dead weight of the tool and hand, offset condition or touch by hand. Once the axis 4 rotates, the axis 3 will move up or down. Take careful precautions not to have your hand and leg caught in it. The axis 4 motor of the TH850A and TH1050A robots is equipped with a brake.

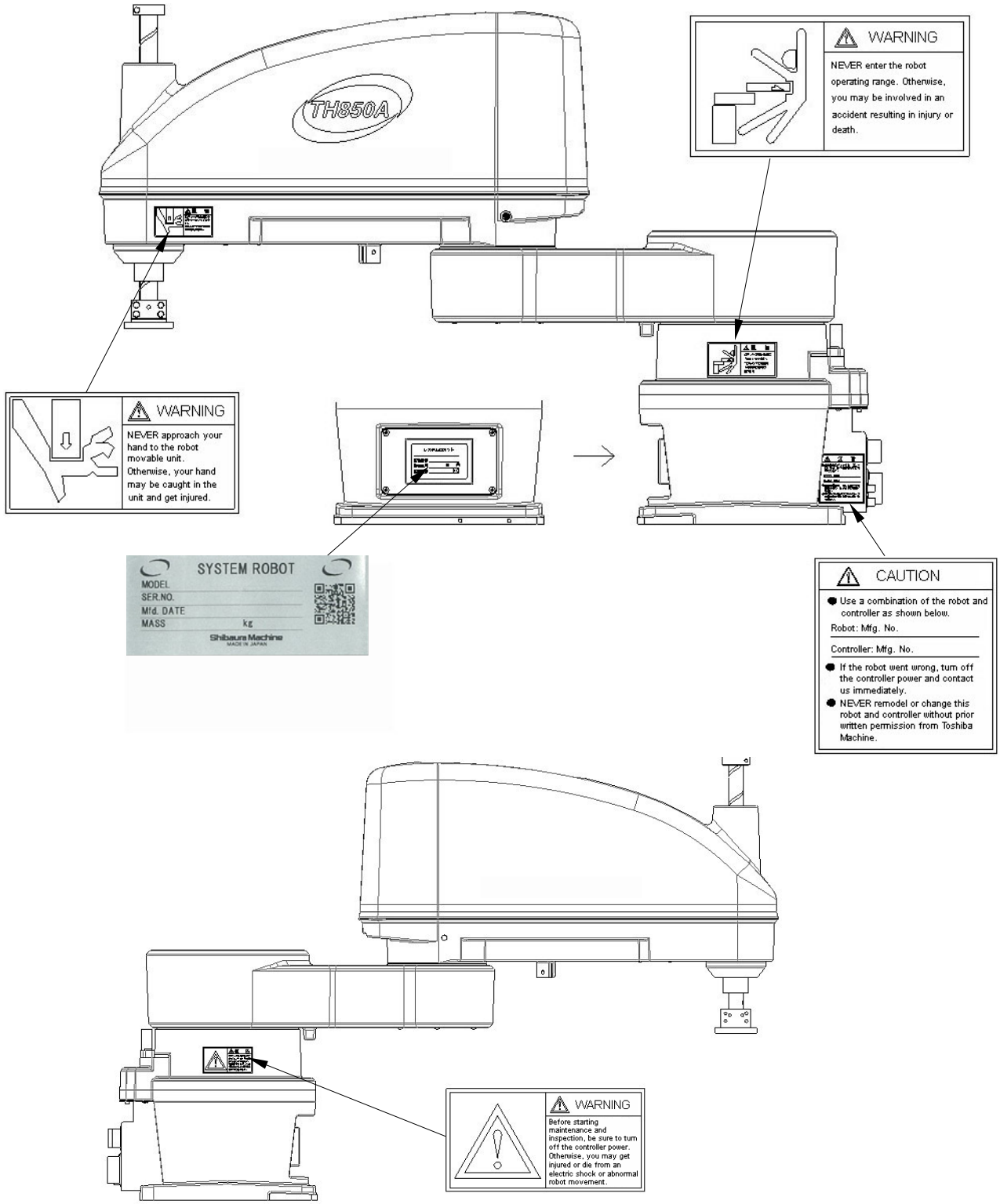
Locations of Warning Labels

(1) Locations of robot warning labels

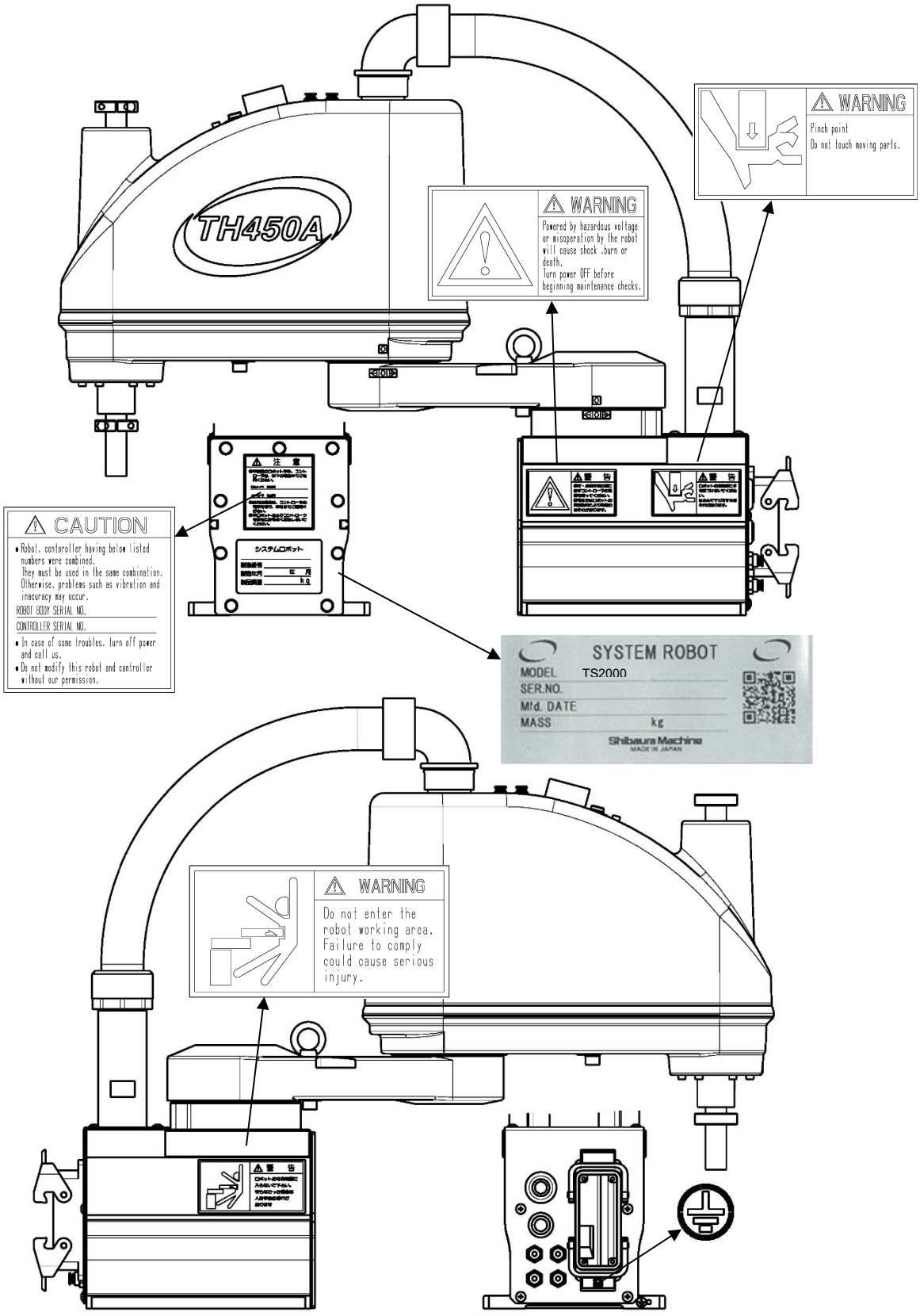
- TH650A



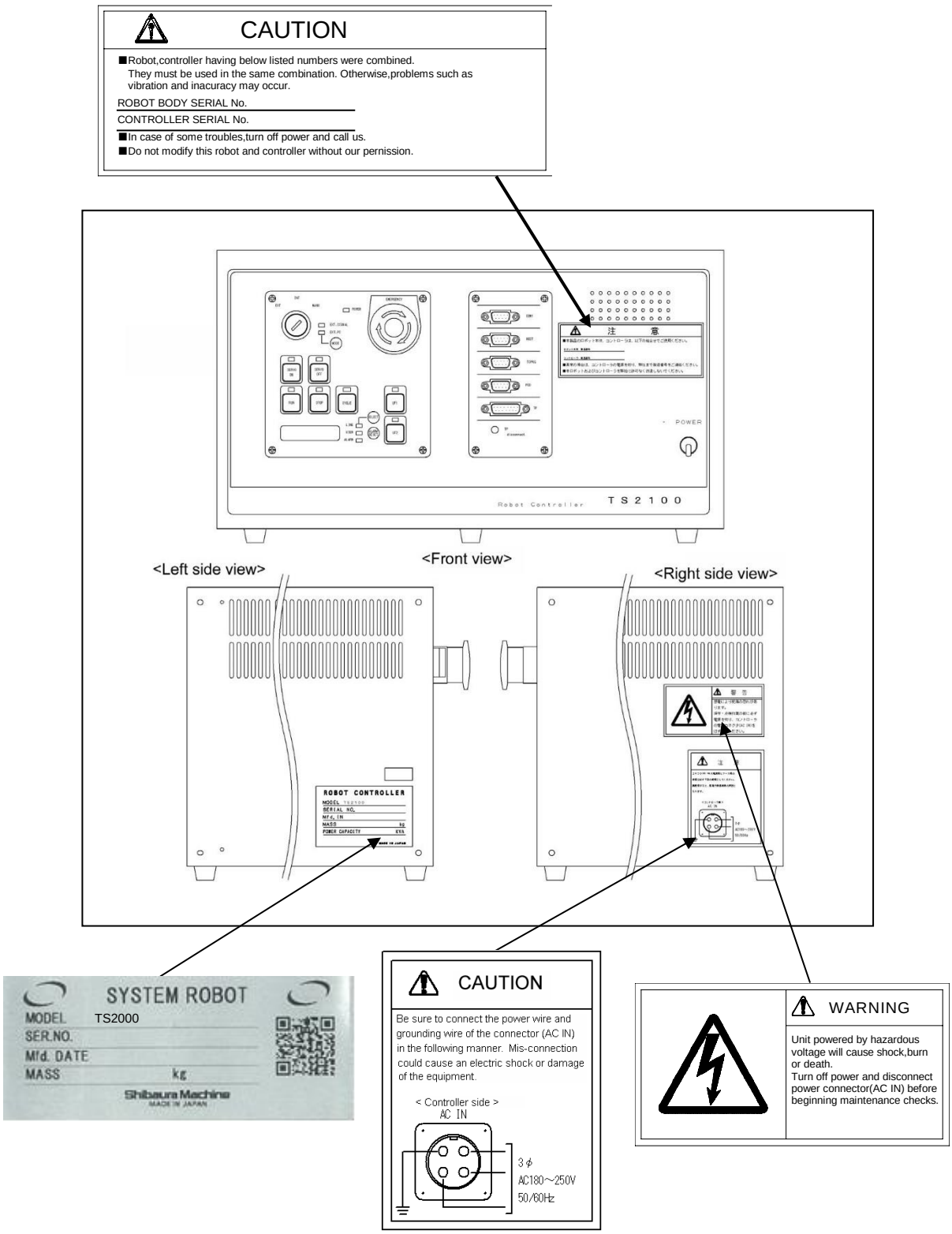
- TH850A, 1050A



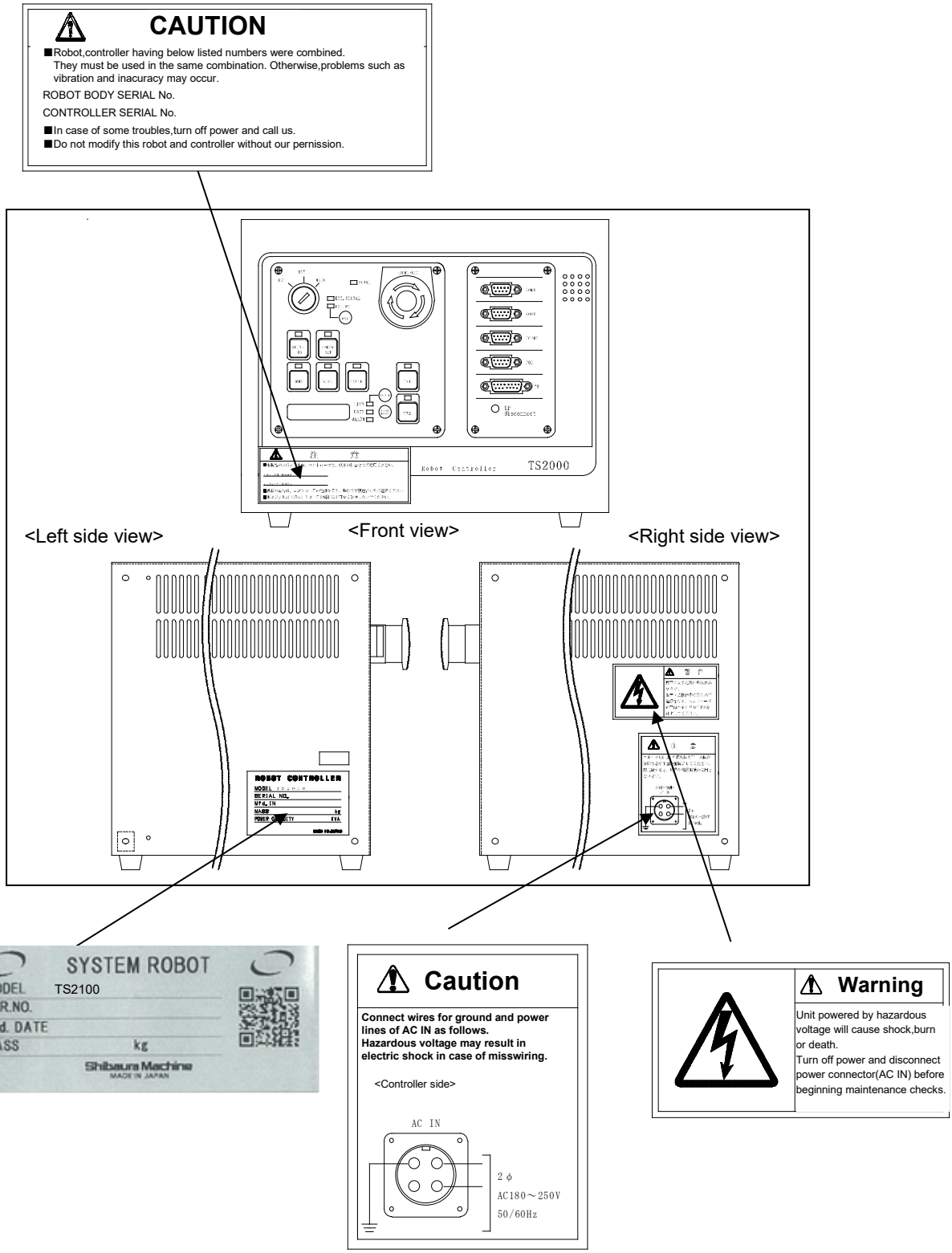
- TH450A, 550A


TH450A
WARNING
Pinch point
Do not touch moving parts.
WARNING
Powered by hazardous voltage or misoperation by the robot will cause shock ,burn or death.
Turn power OFF before beginning maintenance checks.
CAUTION
• Robot, controller having below listed numbers were combined. They must be used in the same combination. Otherwise, problems such as vibration and inaccuracy may occur.
ROBOT BODY SERIAL NO.
CONTROLLER SERIAL NO.
• In case of some troubles, turn off power and call us.
• Do not modify this robot and controller without our permission.
SYSTEM ROBOT
MODEL TS2000
SER.NO.
Mfd. DATE
MASS kg
Shibaura Machine
WARNING
Do not enter the robot working area. Failure to comply could cause serious injury.

(2) Locations of robot warning labels (TS2100)



(3) Locations of robot controller warning labels (TS2000)



Safety Measures

This section describes the necessity of safety measures, safety functions provided for the robot and robot controller, and general safety measures to be taken.

1 Cautions on Safety

The TH series robot and robot controller are equipped with various safety functions. When actually operating the robot, however, the following dangers will be supposed.

a) Danger supposed in normal automatic operation
- Operator's mis-operation and mis-judgment, incomplete program.
- Unexpected robot movement, release or drop of workpiece due to fault of an electronic control device.

b) Danger supposed at teaching and inspection
- Danger of an operator entering the movable range of the robot.
- Operator's negligence from confusion or experience at generation of an unexpected abnormality, and operator's mis-operation due to shortage of behavior and knowledge.
- Approach of an operator to the robot due to unexpected complex movement of the robot.
- Abnormal movement, etc. caused by mis-wiring, contact failure, deterioration and noise.

c) Danger supposed in a related machine, etc.
- Sudden movement of the robot with a command from a related machine, etc.
- Sudden movement of a related machine after the robot movement.
- Danger of an operator being caught or entangled in the robot when teaching, inspecting or adjusting the robot while moving a related machine.

To use the robot safely, safety measures should be taken according to the operating conditions. Otherwise, an unexpected disaster may occur.
If there are safety rules and regulations, strictly observe them. Also, refer to all manuals relating to the robot and robot controller.

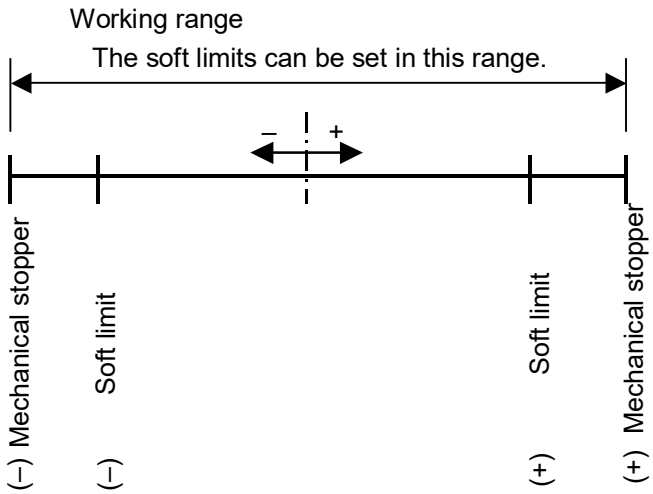
2 Safety Functions

This robot and robot controller have various safety functions as shown below.

a) Self-diagnostic function
This robot has extensive self-diagnostic functions intended for preventing mis-operation of the robot by monitoring faults in the robot and robot controller. Such faults are limited, however, to those that can be detected by the robot controller.

b) Emergency stop function
An EMERGENCY stop pushbutton switch is equipped on the robot controller and teach pendant, respectively. When this switch is pressed, the robot stops immediately. Also, a function effecting emergency stop with an external signal is also available.

c) Overrun prevention function
To prevent overrun, this robot is equipped with soft limits and mechanical stoppers. The mechanical stoppers are provided on axis 1, axis 2 and axis 3, which prevent the moving part of the robot from overrunning.



d) Working range limiting function
Soft limits can be set inside the robot working range. Thus, the working range of the robot can be minimized.

e) Operating speed limiting function
At manual guide and test operation, operating speed of the robot is set at 20 % of the maximum speed to assure easy and safe teaching operation.

f) Operation key on teach pendant
An enable switch is equipped on the teach pendant to enhance safety. When manually guiding the robot, the operator should keep pressing this enable switch while carrying the pendant. Otherwise, the robot will not move even by pressing an appropriate operation key. This enable switch is of a three (3) position-type, and if it is pressed hard, the robot will not move, either.
The robot moves as long as the operation key is pressed. When the switch is released, the robot stops.

g) Master switch on control panel
Key switch used to change over the manual and automatic operation modes.
If an operator has to enter the operating range inevitably for teaching operation or inspection, he should carry a key with him so that anyone cannot change over the mode.

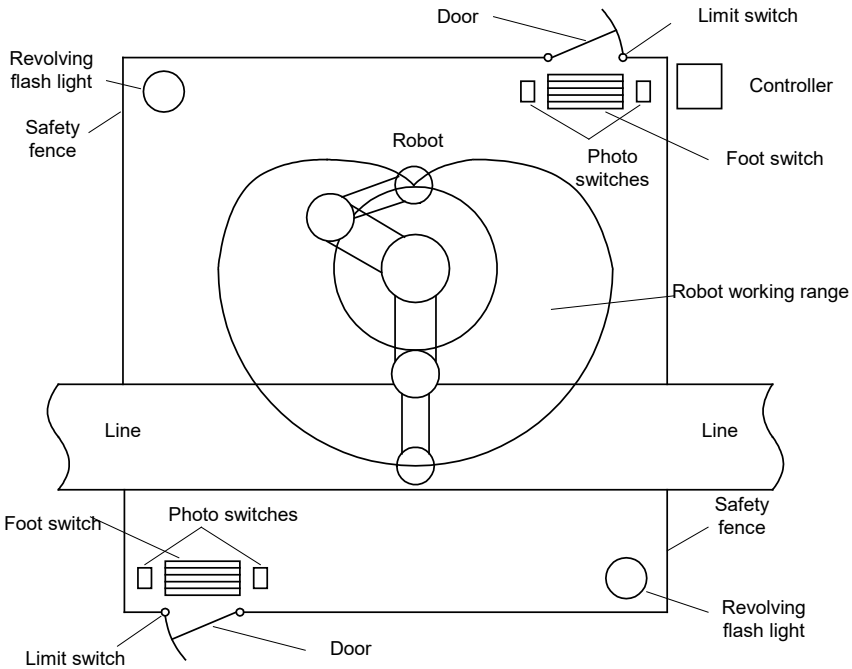
3 Safety Measures

Most of disasters caused by the robot originate from unsafe behavior of man.
When using the robot, he should foresee what will involve danger and try to prevent such a dangerous condition. Operation should be done only after all safety conditions are satisfied.
Main safety measures are as follows:

a) General cautions on using the robot

1) When installing the robot, provide a space required for the job safely.

2) The dangerous area should be identified. For this purpose, necessary measures should be taken to prevent entry of any person, by installing safety fences, etc.



The dangerous area signifies an area near the robot, where, if a person has entered, he will jeopardize.

3) Limit switches, photo switches, foot switch, etc. should be equipped on the doorway of each safety fence so that emergency stop can be effected on the robot should a person enter the dangerous area.

The emergency stop device should be electrically independent, which should be compulsively opened at normal close contact (i.e., contact closed at normal operation) and should not be returned automatically.

4) The robot controller should be installed outside the dangerous area where an operator can watch the robot movements.

5) Operation should be performed only by a well-trained and qualified operator. Anyone who does not understand and is not familiar with the movements of the robot and related equipment should not execute operation. Also, current condition of the robot should always be displayed to prevent an unrelated person from carelessly entering the working range or operating the robot.

6) Before day's operation, perform the following check. Pushbutton switches equipped on the control panel and teach pendant should always be operated while confirmed visually.

<Check before operation>

- Make sure that visual appearance of the robot, controller, peripheral equipment and cables is in good condition.
- Make sure that no obstacle stands in or near the working range of the robot and peripheral equipment.
- Make sure that the emergency stop and other safety devices operate properly.
- Make sure that no abnormal noise or vibration is involved in the robot operation.

b) Cautions on teaching operation

In principle, teaching operation should be performed outside the dangerous area of the robot. If it should be performed inevitably within the dangerous area, strictly observe the following cautions.

1) The teaching operation should always be performed by two (2) persons. One person performs the job and the other person watches outside the dangerous area. Also, both persons should try to prevent mis-operation with each other.

2) The operator should do the job in an attitude ready to press the EMERGENCY stop pushbutton switch at any time. Also, he should perform the job at a position from which he can evacuate immediately at the time of an emergency after confirming the robot operating zone and shields in the surroundings. Also, he should not turn his back to the robot.

3) The supervisor should keep watch on the job at a position where he can see the entire robot system and operate the EMERGENCY stop pushbutton switch at the time of an emergency. Also, he should keep anyone from entering the dangerous area.

c) Other cautions

1) The gripping unit of the robot should not stick out, except for the part required for operation. Also, even at a sudden stop due to power failure, malfunction or emergency stop during operation, the robot should hold a workpiece in a stable posture.

2) If the robot has malfunctioned, turn the power off, identify and remove the cause of the nonconformity, maintain the peripheral equipment and completely restore the malfunctioned robot. Then confirm its movements at a low speed. Even if the robot has stopped, DO NOT approach the dangerous area immediately.

3) Before entering the dangerous area of the robot for inspection, maintenance or repair, be sure to turn the power off. Also, turn the power off when the robot is not in use.